

GradeWise Service

Automated Exam Evaluation using AI

- ✓ Question Paper >
- ✓ Model Answer >
- ✓ Student Sheet >
- 4 Results

Start New Evaluation

Evaluation Result

Student ID: MH001

P
21.5 / 40.0

Scoreboard

Question	Score	Max	%		Status
Section A					
Q1	1	1	100%	1/1	Scored
Q2	1	1	100%	1/1	Scored
Q3	1	1	100%	1/1	Scored
Q4	1	1	100%	1/1	Scored
Q5	1	1	100%	1/1	Scored
Q6	1	1	100%	1/1	Scored
Q7	1	1	100%	1/1	Scored

Question	Score	Max	%		Status
Q8	1	1	100%	1/1	Scored
Section B					
Q9	2	2	100%	2/2	Scored
Q10	0	2	0%	0/2	Not Scored
Q11	1	2	50%	1/2	Scored
Q12	1	2	50%	1/2	Scored
Q13	1	2	50%	1/2	Scored
Q14i	0	1	0%	0/1	Not Scored
Q14ii	0	1	0%	0/1	Not Scored
Section C					
Q15	1.5	3	50%	1.5/3	Scored
Q16i	0	1.5	0%	0/1.5	Not Scored
Q16ii	0	1.5	0%	0/1.5	Not Scored
Q17	1.5	3	50%	1.5/3	Scored
Q18	1.5	3	50%	1.5/3	Scored
Section D					

Question	Score	Max	%	Status
Q19	4	8	50%	4/8 Scored
Total	21.5	40	54%	—

Question-wise Details

Question 1

1 / 1

The student's answer matches the model answer exactly, selecting option (b) 8, which is the correct number of subsets of the set $A = \{1, 2, 3\}$. Therefore, full marks are awarded.

Question 2

1 / 1

The student's answer matches the model answer exactly in meaning and choice, correctly identifying the universal set as the set containing all elements under consideration. Minor differences in capitalization do not affect correctness.

ⓘ Suggestions

- Ensure consistent capitalization to match the model answer format.
- Continue to provide clear and precise answers.

Question 3

1 / 1

The student's answer matches the model answer exactly. Since $\sin A = 3/5$ and A is acute, $\cos A = 4/5$ is correct, corresponding to option (a). Full marks are awarded.

Question 4

1 / 1

The student's answer matches the model answer exactly in meaning and choice, with only a minor difference in spacing around the hyphen which does not affect correctness. Therefore, full marks are awarded.

 Suggestions

- Ensure consistent formatting of terms, such as using en-dashes or hyphens uniformly.
- Maintain clarity by avoiding extra spaces around hyphens in compound terms.

Question 5

1 / 1

The student's answer matches the model answer exactly, both indicating that $\sin 30^\circ$ is (b) $1/2$. Therefore, full marks are awarded.

Question 6

1 / 1

The student's answer matches the model answer exactly, both selecting option (c) 60 as the number of ways to arrange 3 books out of 5 different books. This is the correct answer, so full marks are awarded.

 Suggestions

- No improvements needed as the answer is correct and complete.

Question 7

1 / 1

The student's answer exactly matches the model answer, correctly identifying that if A and B are disjoint sets, then their intersection is the empty set ($A \cap B = \emptyset$).

 Suggestions

- Ensure to read the question carefully and select the option that correctly represents the property of disjoint sets.
- No improvements needed as the answer is correct and complete.

Question 8

1 / 1

The student's answer matches the model answer exactly. The relation 'less than' on the set of natural numbers is transitive, and the student correctly identified

option (c).

🚩 **Suggestions**

- No improvements needed as the answer is correct and complete.

Question 9

2 / 2

The student's answer correctly defines a set as a well-defined collection of objects and provides a valid example $\{1, 2, 3\}$. The answer is complete and accurate, matching the model answer in meaning and content.

🚩 **Suggestions**

- Include a label for the example set, such as ' $A = \{1, 2, 3\}$ ', to improve clarity.
- Add spaces after commas in the example for better readability.

Question 10

0 / 2

The student's answer correctly identifies the domain as all real numbers ($\mathbb{R}$), which matches the model answer. However, the range is incorrect. The function $f(x) = x^2$ produces only non-negative values, so the range should be $\{x : x \geq 0\}$ or $[0, \infty)$. The student's range of $[-\infty, \infty)$ is incorrect because it includes negative values, which are not possible outputs of the function.

🚩 **Suggestions**

- Review the concept of the range of a function, especially for quadratic functions like $f(x) = x^2$.
- Remember that squaring any real number results in a non-negative value, so the range should start from 0 and extend to positive infinity.
- Use interval notation correctly to express ranges; for example, $[0, \infty)$ represents all real numbers from 0 to infinity, including 0.

Question 11

1 / 2

The student provided the correct final numerical answer of $1/2$ but did not show any working or steps to arrive at this result. According to the grading rules, this warrants half marks.

⚠ Suggestions

- Show the intermediate steps, such as writing $\sin 45^\circ = 1/\sqrt{2}$ and $\cos 45^\circ = 1/\sqrt{2}$ before multiplying.
- Use consistent notation and spacing for trigonometric functions and angles, e.g., $\sin 45^\circ$ instead of $\text{Sin}45$.
- Include the multiplication step explicitly to demonstrate understanding.

Question 12

1 / 2

The student provided the correct final numerical value of 20 for 5P_4 , which matches the model answer. However, the student did not show any steps or working to arrive at this answer, and the notation used ('\$P_1') is incorrect for the permutation expression. According to grading rules, correct final answer without steps merits half marks.

⚠ Suggestions

- Show the calculation steps to demonstrate understanding, e.g., $5 \times 4 \times 3 \times 2 = 120$ for 5P_4 .
- Use correct notation for permutations, such as 5P_4 or $P(5,4)$, instead of '\$P_1'.
- Include intermediate steps to improve clarity and completeness.

Question 13

1 / 2

The student provided the correct final numerical answer (16) for the number of elements in the power set of A, which matches the model answer. However, the student did not show any steps or reasoning to arrive at this answer. According to the grading rules, this warrants half marks.

⚠ Suggestions

- Include the formula or reasoning used to find the number of elements in the power set, such as stating that the power set of a set with n elements has 2^n elements.
- Show the calculation explicitly, for example: $2^4 = 16$.

Question 14i

0 / 1

No student answer available.

🚫 **Suggestions**

- Provide an answer to receive credit.

Question 14ii

0 / 1

No student answer available.

🚫 **Suggestions**

- Provide an answer to receive credit.

Question 15

1.5 / 3

The student has provided the correct final conclusion that the identity is proved, but has not shown any algebraic steps or working. According to the grading rules, a correct final answer without steps merits half the maximum score. The student correctly identifies the identity as proved but lacks the demonstration of the proof process.

🚫 **Suggestions**

- Show the algebraic manipulation step-by-step to demonstrate how the left-hand side simplifies to the right-hand side.
- Include intermediate steps such as combining fractions, expanding squares, and using trigonometric identities.
- Avoid skipping steps to ensure full marks and to clearly communicate your reasoning.

Question 16i

0 / 1.5

No student answer available.

🚫 **Suggestions**

- Provide an answer to receive credit.

Question 16ii**0 / 1.5**

No student answer available.

🚩 Suggestions

- Provide an answer to receive credit.

Question 17**1.5 / 3**

The student provided the correct final numerical answer (60) but did not show the calculation steps or reasoning. The notation ' 5P_3 ' is incomplete and unclear, but since the numerical value matches the model answer, half marks are awarded.

🚩 Suggestions

- Show the full calculation steps, such as $5 \times 4 \times 3 = 60$, to demonstrate understanding.
- Use proper notation for permutations, for example, 5P_3 or write 'Number of 3-digit numbers = $5 \times 4 \times 3 = 60$ '.
- Explain that digits cannot be repeated and that the first digit cannot be zero (if applicable).

Question 18**1.5 / 3**

The student correctly identified that the relation is reflexive, symmetric, and transitive, which matches the model answer's conclusion. However, the student did not provide any reasoning or explanation to support these claims. According to the grading rules, a correct final answer without supporting steps or justification merits half the maximum score.

🚩 Suggestions

- Provide reasoning or examples to show why the relation is reflexive, symmetric, and transitive.
- Explain that since all elements (a,a) are in R for all a in A , the relation is reflexive.
- Demonstrate that for every (a,b) in R , (b,a) is also in R to show symmetry.
- Show that for any (a,b) and (b,c) in R , (a,c) is in R to prove transitivity.
- Including these explanations will improve completeness and clarity.

Question 19

4 / 8

The student correctly states the identity and concludes that it is proved, matching the model answer's final expression. However, the student provides only a brief explanation without showing the intermediate step of rewriting $(1 + \tan^2\theta)$ as $\sec^2\theta$, which is crucial for a complete proof. Therefore, the answer is correct but lacks the detailed working required for full marks.

🚩 Suggestions

- Include the key trigonometric identity $\tan^2\theta = \sec^2\theta - 1$ and show the substitution step explicitly.
- Demonstrate the simplification from LHS to RHS step-by-step to strengthen the proof.
- Avoid just stating the identity is proved; provide clear reasoning to support the conclusion.